

Chapter One

Kirchhoff's Law

Basic Concepts

Some books are to be tasted, others to be swallowed, and some few to be chewed and digested.

—Francis Bacon

1.1 Introduction

Electric circuit theory and electromagnetic theory are the two fundamental theories upon which all branches of electrical engineering are built. Many branches of electrical engineering, such as power, electric machines, control, electronics, communications, and instrumentation, are based on electric circuit theory. Therefore, the basic electric circuit theory course is the most important course for an electrical engineering student, and always an excellent starting point for a beginning student in electrical engineering education. Circuit theory is also valuable to students specializing in other branches of the physical sciences because circuits are a good model for the study of energy systems in general, and because of the applied mathematics, physics, and topology involved.

In electrical engineering, we are often interested in communicating or transferring energy from one point to another. To do this requires an interconnection of electrical devices. Such interconnection is referred to as an electric circuit, and each component of the circuit is known as an element.

An **electric circuit** is an interconnection of electrical elements.

A simple electric circuit is shown in Fig. 1.1. It consists of three basic elements: a battery, a lamp, and connecting wires. Such a simple circuit can exist by itself; it has several applications, such as a flashlight, a search light, and so forth.

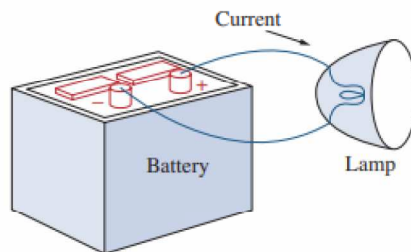


Figure 1.1: A simple electric circuit

Charge

Charge is an electrical property of the atomic particles of which matter consists, measured in coulombs (C).

We know from elementary physics that all matter is made of fundamental building blocks known as atoms and that each atom consists of electrons, protons, and neutrons. We also know that the charge e on an electron is negative and equal in magnitude to C , while a proton carries a positive charge of the same magnitude as the electron. The presence of equal numbers of protons and electrons leaves an atom neutrally charged.

The following points should be noted about electric charge:

- The coulomb is a large unit for charges. In 1 C of charge, there are $\frac{1}{1.602 \times 10^{-19}} = 6.24 \times 10^{18}$ electrons. Thus realistic or laboratory values of charges are on the order of pC, nC, or μ C.
- According to experimental observations, the only charges that occur in nature are integral multiples of the electronic charge $e = -1.602 \times 10^{-19} C$.
- The law of conservation of charge states that charge can neither be created nor destroyed only transferred. Thus the algebraic sum of the electric charges in a system does not change.

Example: How much charge is represented by 4,600 electrons?

Solution: Each electron has charge $-1.602 \times 10^{-19} \text{ C}$. Hence 4,600 electrons will have $-1.602 \times 10^{-19} \text{ C}$ C/electron $\times$ 4,600 electrons = $-7.369 \times 10^{-16} \text{ C}$

Electric current

Electric current is the time rate of change of charge, measured in amperes (A).

When a conducting wire (consisting of several atoms) is connected to a battery (a source of electromotive force), the charges are compelled to move; positive charges move in one direction while negative charges move in the opposite direction. This motion of charges creates electric current. It is conventional to take the current flow as the movement of positive charges. That is, opposite to the flow of negative charges, as Fig. 1.2 illustrates. This convention was introduced by Benjamin Franklin (1706–1790), the American scientist and inventor. Although we now know that current in metallic conductors is due to negatively charged electrons, we will follow the universally accepted convention that current is the net flow of positive charges.

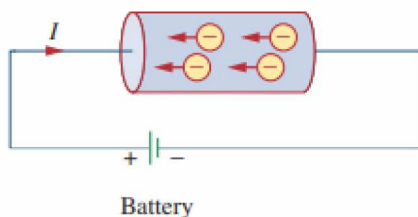


Figure 1.2: Electric current due to flow of electronic charge in a conductor.

Mathematically, the relationship between current I , charge Q , and time t is

$$I = \frac{dQ}{dt}$$

$$Q = \int I dt$$

Or

where current is measured in amperes (A), and

$$1 \text{ amperes} = 1 \frac{\text{coulomb}}{\text{second}}$$

If the current does not change with time, but remains constant, we call it a direct current (dc).

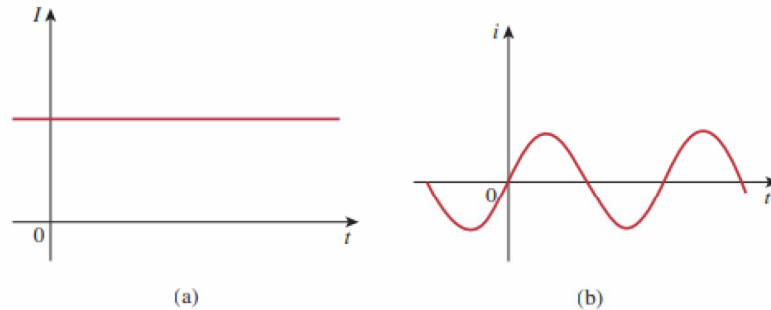


Figure 1.3: Two common types of current: (a) direct current (dc), (b) alternating current (ac).

By convention the symbol I is used to represent such a constant current.

An alternating current (ac) is a current that varies sinusoidally with time.

A time-varying current is represented by the symbol i . A common form of time-varying current is the sinusoidal current or alternating current (ac). Such current is used in your household to run the air conditioner, refrigerator, washing machine, and other electric appliances.

Voltage

To move the electron in a conductor in a particular direction requires some work or energy transfer. This work is performed by an external electromotive force (emf), typically represented by the battery. This emf is also known as voltage or potential difference. The voltage between two points a and b in an electric circuit is the energy (or work) needed to move a unit charge from a to b ; mathematically,

$$V_{ab} = \frac{dW}{dq}$$

Where w is energy in joules (J) and q is charge in coulombs (C). The voltage or simply V is measured in volts (V), named in honor of the Italian physicist Alessandro Antonio Volta (1745–1827), who invented the first voltaic battery. From Eq. (1.3), it is evident that

$$1 \text{ Volt} = 1 \frac{\text{joule}}{\text{coulomb}} = 1 \text{ newton} - \frac{\text{meter}}{\text{coulomb}}$$

Thus, Voltage (or potential difference) is the energy required to move a unit charge through an element, measured in volts (V).

Figure shows the voltage across an element (represented by a rectangular block) connected to points a and b . The plus and minus signs are used to define reference direction or voltage polarity. The V_{ab} can be interpreted in two ways: (1) Point a is at a potential of V_{ab} volts higher than point b , or (2) the potential at point a with respect to point b is V_{ab} .

It follows logically that in general

$$V_{ab} = -V_{ba}$$

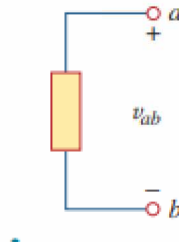


Figure 1.4: Polarity of voltage v_{ab} .

For example, in Figure 1.5, we have two representations of the same voltage. In Fig. 1.5(a), point a is +9V above point b ; in Fig. 1.5(b), point b is -9V above point a . We may say that in Fig. 1.5(a), there is a 9-V voltage drop from a to b or equivalently a 9-V voltage rise from b to a . In other words, a voltage drop from a to b is equivalent to a voltage rise from b to a .

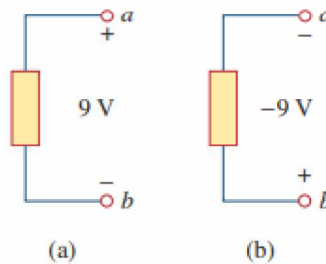


Figure 1.5: Two equivalent representations of the same voltage: (a) Point a is 9 V above point b ; (b) point b is -9 V above point a .

Current and voltage are the two basic variables in electric circuits. The common term signal is used for an electric quantity such as a current or a voltage (or even electromagnetic wave) when it is used for conveying information. Engineers prefer to call such variables signals rather than mathematical functions of time because of their importance in communications and other disciplines. Like electric current, a constant voltage is called a dc voltage and is represented by V , whereas a sinusoidally time-varying voltage is called an ac voltage and is represented by v . A dc voltage is commonly produced by a battery; ac voltage is produced by an electric generator. Power is the time rate of expending or absorbing energy, measured in watts (W).

We write this relationship as

$$P = \frac{dw}{dt}$$

where p is power in watts (W), w is energy in joules (J), and t is time in seconds (s).

We can write it as

$$P = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = VI$$

$$P = VI$$

the energy absorbed or supplied is given by

$$w = \int p \, dt = \int VI \, dt$$

Energy is the capacity to do work, measured in joules (J). The electric power utility companies measure energy in watt-hours (Wh), where

$$1 \text{ Wh} = 3,600 \text{ J}$$

Circuit Elements

An electric circuit is simply an interconnection of the elements. Circuit analysis is the process of determining voltages across (or the currents through) the elements of the circuit.

There are two types of elements found in electric circuits: **Passive elements and active elements**. An **active element** is capable of generating energy while a passive element is not. Examples of **passive elements** are resistors, capacitors, and inductors. Typical active elements include generators, batteries, and operational amplifiers.

Active elements

The most important active elements are voltage or current sources that generally deliver power to the circuit connected to them. There are two kinds of sources: independent and dependent sources.

An **ideal independent source** is an active element that provides a specified voltage or current that is completely independent of other circuit elements.

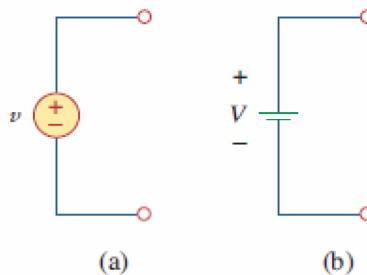


Figure 1.6: Symbols for independent voltage sources:

(a) used for constant or time-varying voltage,

(b) used for constant voltage (dc).

In other words, an ideal independent voltage source delivers to the circuit whatever current is necessary to maintain its terminal voltage. Physical sources such as batteries and generators may be regarded as approximations to ideal voltage sources.

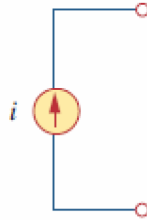


Figure 1.7: Symbol for independent current source.

Similarly, an ideal independent current source is an active element that provides a specified current completely independent of the voltage across the source. That is, the current source delivers to the circuit whatever voltage is necessary to maintain the designated current.

An ideal dependent (or controlled) source is an active element in which the source quantity is controlled by another voltage or current.

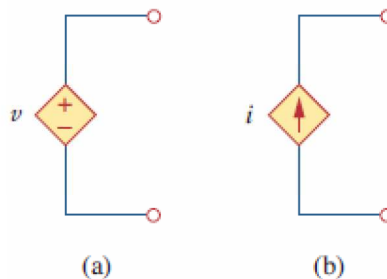


Figure 1.8: Symbols for: (a) dependent voltage source, (b) dependent current source.

Dependent sources are usually designated by diamond-shaped symbols. Since the control of the dependent source is achieved by a voltage or current of some other element in the circuit, and the source can be voltage or current, it follows that there are four possible types of dependent sources, namely:

1. A voltage-controlled voltage source (VCVS).
2. A current-controlled voltage source (CCVS).
3. A voltage-controlled current source (VCCS).
4. A current-controlled current source (CCCS).

Dependent sources are useful in modeling elements such as transistors, operational amplifiers, and integrated circuits. An example of a current-controlled voltage source is shown Figure 1.9

where the voltage $10i$ of the voltage source depends on the current i through element C . Students might be surprised that the value of the dependent voltage source is $10iV$ (and not $10iA$) because it is a voltage source. The key idea to keep in mind is that a voltage source comes with polarities (+ -) in its symbol, while a current source comes with an arrow, irrespective of what it depends on.

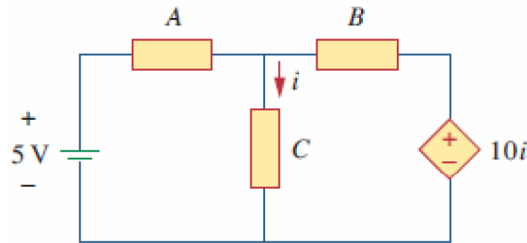


Figure 1.9: The source on the right-hand side is a current-controlled voltage source.

1.2 Basic Laws

Ohm's Law

Ohm's law states that the voltage v across a resistor is directly proportional to the current i flowing through the resistor.

That is,

$$V \propto I$$

Ohm defined the constant of proportionality for a resistor to be the resistance; R (ohms). (The resistance is material property which can change if the internal or external conditions of the element are altered, e.g., if there are changes in the temperature.)

$$V = RI$$

For other circuit elements such as L & C , we have the following relationships between voltage v and current i ,

For the inductor,

$$V = L \frac{dI}{dt} \quad \text{or} \quad I = \frac{1}{L} \int V dt$$

where L is its inductance in henrys (H).

For the capacitor,

$$I = C \frac{dV}{dt} \quad \text{or} \quad V = \frac{1}{C} \int I dt$$

Where C is its capacitance in farads (F).

Nodes, Branches, and Loops

A branch represents a single element such as a voltage source or a resistor.

In other words, a branch represents any two-terminal element. The circuit in Fig. 2.10 has five branches, namely, the 10-V voltage source, the 2-A current source, and the three resistors

A node is the point of connection between two or more branches

A node is usually indicated by a dot in a circuit. If a short circuit (a connecting wire) connects two nodes, the two nodes constitute a single node. The circuit in Fig. 2.10 has three nodes, a , b , and c . Notice that the three points that form node b are connected by perfectly conducting wires and therefore constitute a single point. The same is true of the four points forming node c . We demonstrate that the circuit in Fig. 2.10 has only three nodes by redrawing the circuit in Fig. 2.11. The two circuits in Figs. 2.10 and 2.11 are identical. However, for the sake of clarity, nodes b and c are spread out with perfect conductors as in Fig. 2.10.

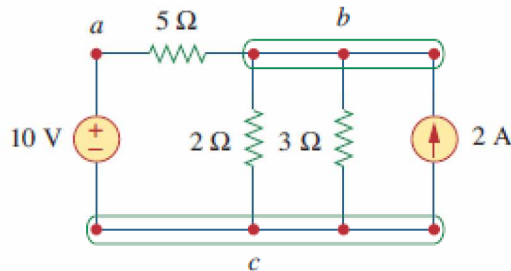


Figure 1.10: Nodes, branches, and loops

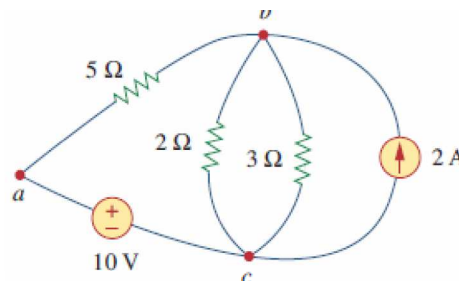


Figure 1.11: The three-node circuit of Fig. 1.10 is redrawn

A loop is any closed path in a circuit.

A loop is a closed path formed by starting at a node, passing through a set of nodes, and returning to the starting node without passing through any node more than once. A loop is said to be *independent* if it contains at least one branch which is not a part of any other independent loop. Independent loops or paths result in independent sets of equations. It is possible to form an independent set of loops where one of the loops does not contain such a branch. In Figure 1.11, $abca$ with the resistor is independent. A second loop with the resistor and the current source is independent. The third loop could be the one with the resistor in parallel with the resistor. This does form an independent set of loops.

A network with b branches, n nodes, and l independent loops will satisfy the fundamental theorem of network topology:

$$b = l + n - 1$$

Kirchhoff's Laws

Ohm's law by itself is not sufficient to analyze circuits. However, when it is coupled with Kirchhoff's two laws, we have a sufficient, powerful set of tools for analyzing a large variety of electric circuits. Kirchhoff's laws were first introduced in 1847 by the German physicist Gustav Robert Kirchhoff (1824–1887). These laws are formally known as Kirchhoff's current law (KCL) and Kirchhoff's voltage law (KVL).

Kirchhoff's first law is based on the law of conservation of charge, which requires that the algebraic sum of charges within a system cannot change.

Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around any closed loop of a circuit is zero.

It is expressed mathematically as

$$\sum_{k=1}^n V_k = 0$$

where n is the number of voltages in the loop (or the number of branches in the loop) and V_k is the k th voltage.

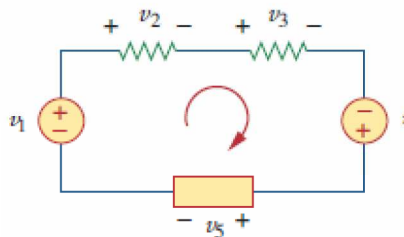


Figure 1.12: A single-loop circuit illustrating KVL

To illustrate KVL, consider the circuit in Fig. 2.19. The sign on each voltage is the polarity of the terminal encountered first as we travel around the loop. We can start with any branch and go around the loop either clockwise or counterclockwise. Suppose we start with the voltage source and go clockwise around the loop as shown; then voltages would be $-v_1, +v_2, +v_3, -v_4$ and $+v_5$ in that order. For example, as we reach branch 3, the positive terminal is met first; hence, we have $+v_3$. For branch 4, we reach the negative terminal first; hence, $-v_4$. Thus, KVL yields

$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

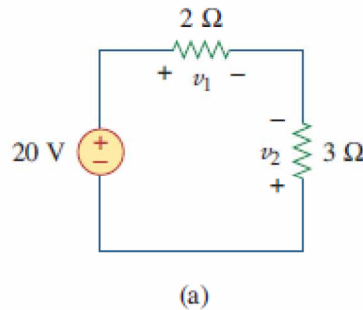
Rearranging terms gives $v_2 + v_3 + v_5 = v_1 + v_4$

which may be interpreted as

Sum of voltage drops = Sum of voltage rises

This is an alternative form of KVL.

Example: For the circuit shown in following fig, find voltages v_1 and v_2 .



Solution: Assume that current i flows through the loop as shown in the figures.

From Ohm's law,

$$v_1 = 2i, \quad v_2 = -3i$$

Applying KVL around the loop gives

$$-20 + v_1 - v_2 = 0$$

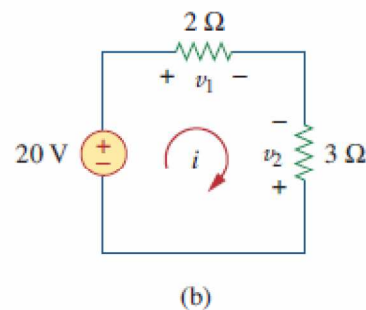
$$-20 + 2i + 3i = 0$$

$$5i = 20$$

$$i = 4\text{A}$$

From this we get,

$$v_1 = 8\text{V}, \quad v_2 = -12\text{V}$$



Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

Mathematically, KCL implies that

$$\sum_{k=1}^m I_k = 0$$

where m is the number of branches connected to the node and I_k is the k th current entering (or leaving) the node.

By this law, currents entering a node may be regarded as positive, while currents leaving the node may be taken as negative or vice versa.

To prove KCL, assume a set of currents flow into a node. The algebraic sum of currents at the node is

$$I_T(t) = I_1(t) + I_2(t) + I_3(t) \dots \dots \dots$$

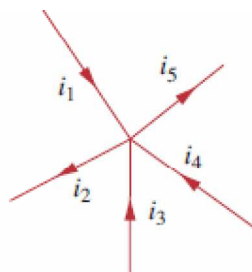
Integrating both sides, we get

$$q_T(t) = q_1(t) + q_2(t) + q_3(t) \dots \dots \dots$$

But the law of conservation of electric charge requires that the algebraic sum of electric charges at the node must not change; that is, the node stores no net charge. Thus

$$q_T(t) = 0 \rightarrow I_T(t) = 0 \text{ confirming the validity of KCL.}$$

Consider the node in Fig. 2.16. Applying KCL gives



$$I_1 - I_2 + I_3 + I_4 - I_5 = 0$$

Figure 1.13: Currents at a node illustrating KCL

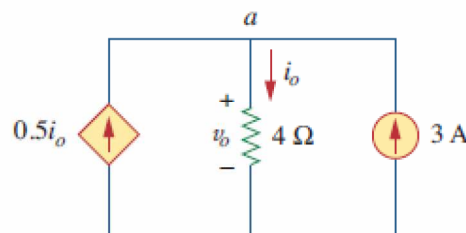
since currents I_1, I_2 and I_4 are entering the node, while currents I_3 and I_5 are leaving it. By rearranging the terms, we get

$$I_1 + I_3 + I_4 = I_2 + I_5$$

Equation (2.17) is an alternative form of KCL:

The sum of the currents entering a node is equal to the sum of the currents leaving the node.

Example: Find current i_o and voltage v_o for the circuit shown in the Fig.



Solution: Applying KCL to node a , we obtain

$$0.5i_o + 3 = i_o \Rightarrow i_o = 6A$$

For the 4- resistor, Ohm's law gives

$$v_o = 4i_o = 24 V$$

Voltage Division Rule

Voltage Division Rule states that the total voltage applied across a series connection of multiple resistors is divided among the resistors in proportional to their resistance. This means, the voltage drop will be maximum across the resistor having maximum value of resistance. Likewise, it will be minimum for resistor having lowest value of resistance.

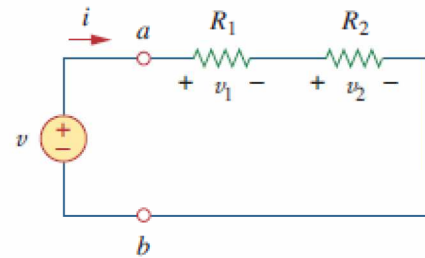


Figure 1.14: A single-loop circuit with two resistors in series

Applying Ohm's law to each of the resistors, we obtain

$$v_1 = iR_1 \quad \dots\dots\dots(a)$$

$$v_2 = iR_2 \quad \dots\dots\dots(b)$$

If we apply KVL to the loop (moving in the clockwise direction), we have

$$-v + v_1 + v_2 = 0$$

$$v = v_1 + v_2$$

Using (a), (b) and (c), we can find current i

$$i = \frac{v}{R_1 + R_2},$$

Putting the value of i , we get

$$v_1 = \left(\frac{v}{R_1 + R_2} \right) R_1$$

$$v_2 = \left(\frac{v}{R_1 + R_2} \right) R_2$$

Current Division Rule

Consider the circuit in the following Fig where two resistors are connected in parallel and therefore have the same voltage across them.

From Ohm's law

$$v = i_1 R_1 = i_2 R_2$$

Or

$$i_1 = \frac{v}{R_1}, \quad i_2 = \frac{v}{R_2}$$

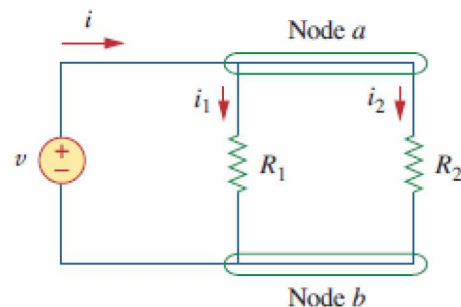
Two resistors in parallel.

Applying KCL at node a gives the total current i as

$$i = i_1 + i_2$$

Solving the equation, we get

$$v = i \left(\frac{R_1 R_2}{R_1 + R_2} \right)$$



This gives,

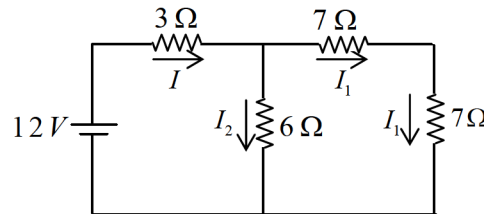
$$i_1 = i \left(\frac{R_2}{R_1 + R_2} \right)$$

$$i_2 = i \left(\frac{R_1}{R_1 + R_2} \right)$$

Example: Find current across each element in the circuit shown in figure.

(a) Using current division rule

(b) Using KVL



Solution:

(a) Current
$$I = \frac{12}{3 + \frac{6 \times 14}{6 + 14}} = \frac{5}{3} A$$

$$I_1 = \frac{6}{6 + 14} \times \frac{5}{3} = \frac{1}{2} A$$

$$I_2 = \frac{14}{6 + 14} \times \frac{5}{3} = \frac{7}{6} A$$

(b) Apply KVL in Loop I

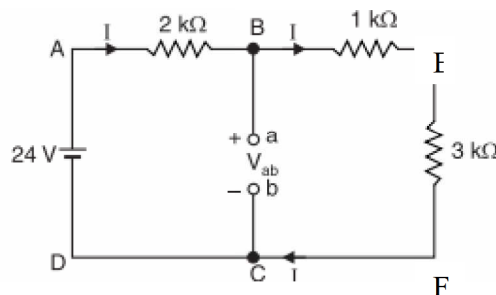
$$-12 + 3(I_1 + I_2) + 6I_2 = 0 \Rightarrow 3I_1 + 9I_2 = 12$$

Apply KVL in Loop II

$$7I_1 + 7I_1 - 6I_2 = 0 \Rightarrow 7I_1 = 3I_2$$

Solving above equations, we get $I_1 = \frac{1}{2} A$

Example: Use Kirchhoff's voltage law to find the voltage V_{ab} in given circuit

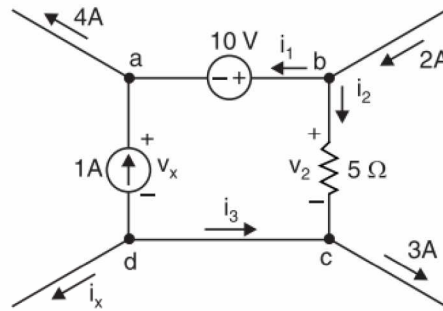


Solutions: Applying KVL in loop AEFDA

$$-24 + 2I + I + 3I = 0 \Rightarrow I = \frac{24}{6} = 4 \text{ mA}$$

$$\text{Voltage across AB } V_{ab} = 4 \times (1 + 3) = 16 \text{ V}$$

Example: Find i_x and v_x in the network shown in given Fig.



Solution: Applying KCL at node **a**

$$1 + i_1 = 4$$

$$i_1 = 3A$$

Applying KCL at node **b**

$$3 + i_2 = 2$$

$$i_2 = -1A$$

Applying KCL at node **c**

$$-1 + i_3 = 3$$

$$i_3 = 4A$$

Applying KCL at node **d**

$$1 + i_x + i_3 = 0$$

Or

$$1 + i_x + 4 = 0$$

$$\Rightarrow i_x = -5A$$

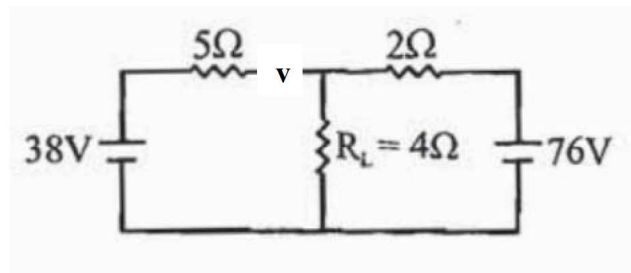
Applying KVL in abcda loop

$$-v_x - 10 + 5 i_2 = 0$$

$$-v_x - 10 - 5 = 0$$

$$\Rightarrow v_x = -15V$$

Example: For the circuit shown, the potential difference (in Volts) across R_L is



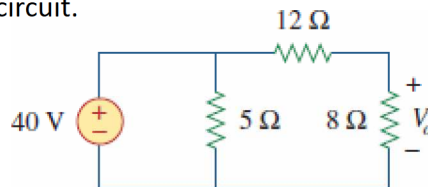
Solution: Let the potential difference across R_L is v

Applying KCL at node v

$$\frac{v - 38}{5} + \frac{v}{4} + \frac{v - 76}{2} = 0$$

Or $19V = 912 \Rightarrow V = 48V$

Example: Find the value of V_0 in given circuit.



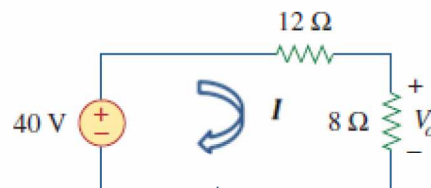
Solution:

Method-I: In the circuit 5Ω resistor can be neglected without any change to load.

Applying KVL,

$$I = \frac{40}{20} = 2 \text{ A}$$

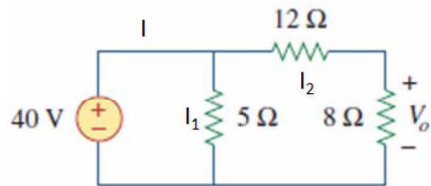
$$V_0 = 16 \text{ V}$$



Method-II:

Let the total current in circuit is I .

$$I = \frac{\text{total voltage}}{\text{total resistance}} = \frac{40}{\frac{5 \times 20}{25}} = 10$$



By current division rule current I_2 in 8Ω resistor is equal to

$$I_2 = \left(\frac{5}{5 + 20} \right) \times 10 = 2 \text{ A}$$

Hence voltage $V_0 = 16 \text{ V}$

Example: Calculate values of V_1 and V_2 across 4Ω and 5Ω resistor in the following figure by KVL

Solution: Applying KVL

$$-6 - 3 - 1 + 5 + 4I + 2I + 2 + 5I = 0$$

$$-10 + 7 + 11I = 0$$

$$I = \frac{3}{11} A$$

$$V_1 = 4I = 4 \times \frac{3}{11} = \frac{12}{11} V$$

$$V_2 = 5I = 5 \times \frac{3}{11} = \frac{15}{11} V$$

